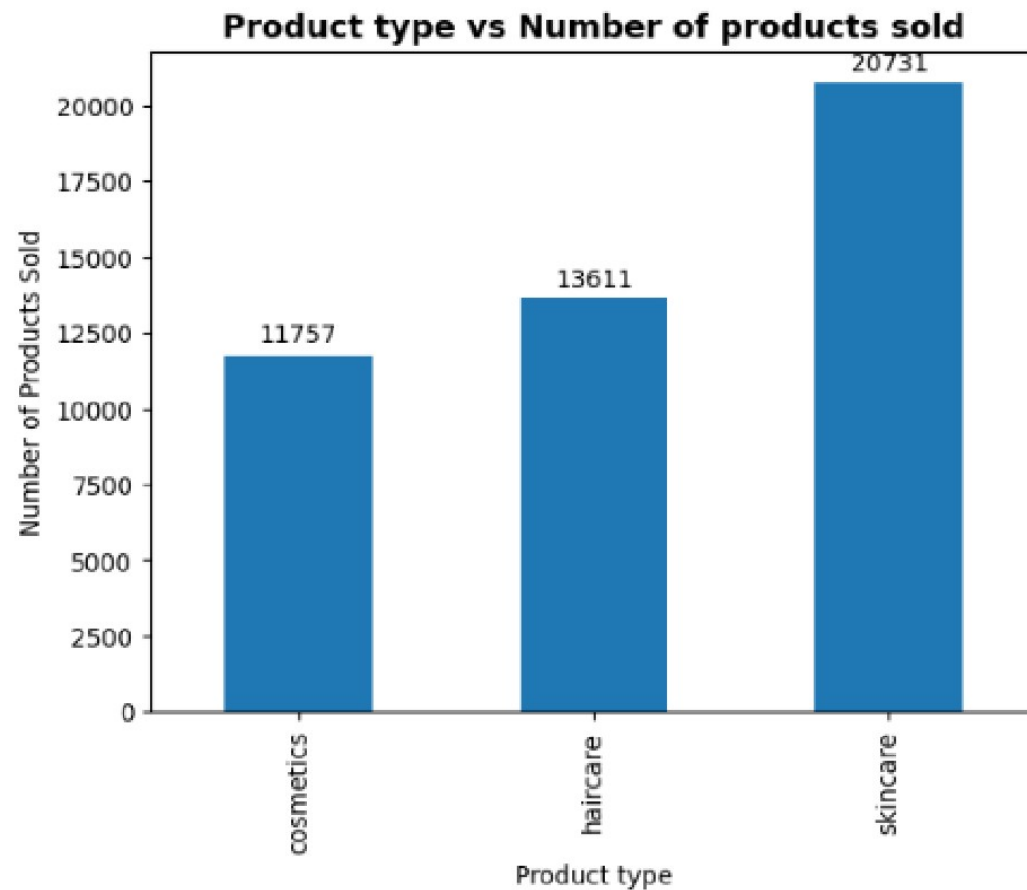


```
[58]: ax = df.groupby("Product type")["Number of products sold"].sum().plot.bar()

# Add numbers on top of the bars
ax.bar_label(ax.containers[0], fmt="%d", padding=3)

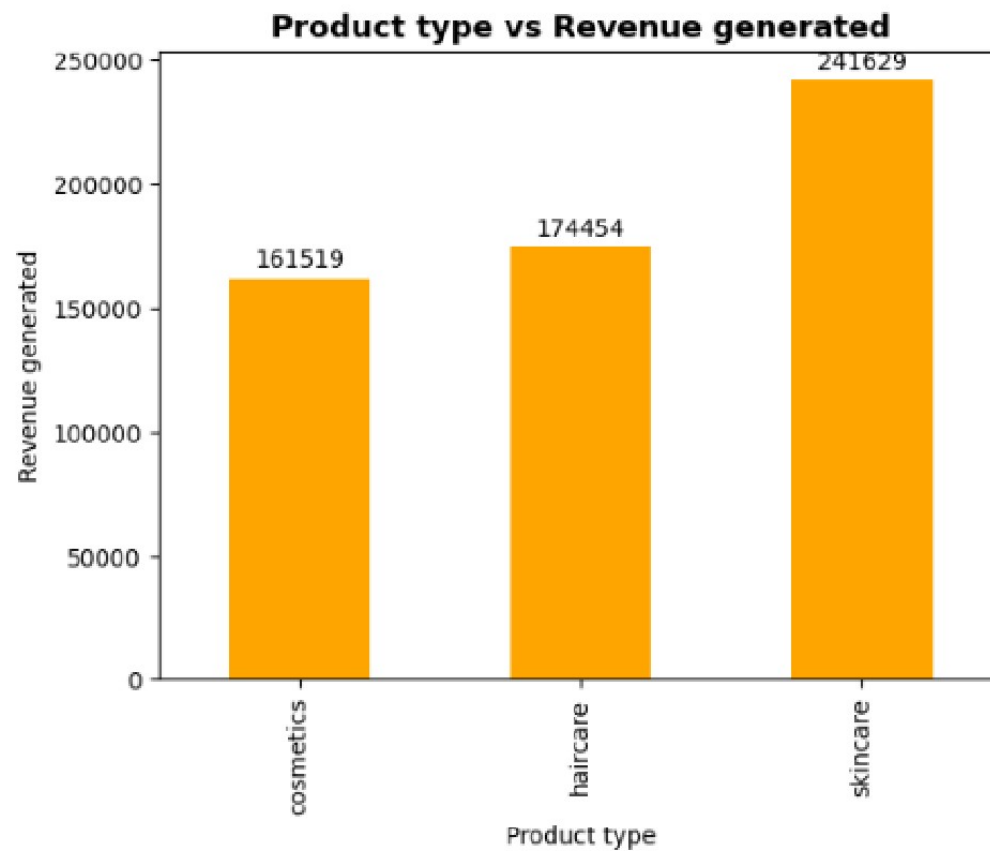
plt.title("Product type vs Number of products sold", fontsize=13, fontweight="bold")
plt.xlabel("Product type")
plt.ylabel("Number of Products Sold")
plt.show()
```



```
[64]: ax = df.groupby("Product type")["Revenue generated"].sum().plot.bar(color="Orange")

# Add numbers on top of the bars
ax.bar_label(ax.containers[0], fmt="%d", padding=3)

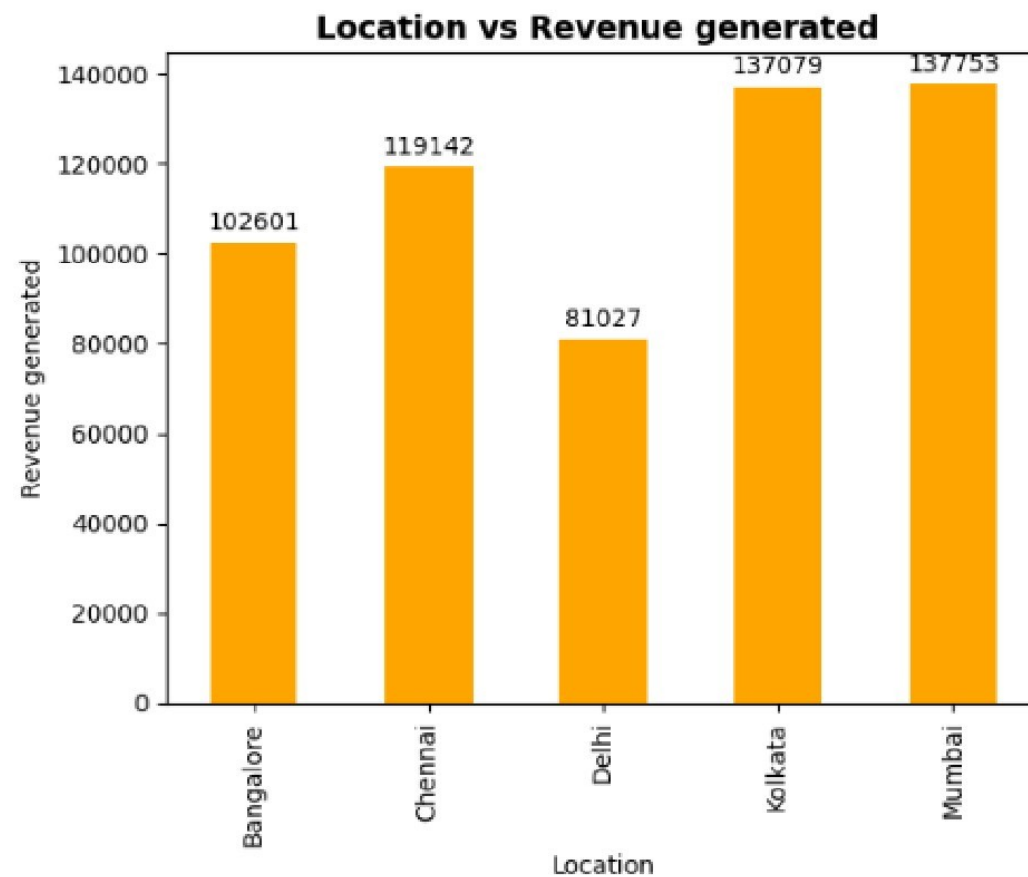
plt.title("Product type vs Revenue generated", fontsize=13, fontweight="bold")
plt.xlabel("Product type")
plt.ylabel("Revenue generated")
plt.show()
```



```
[72]: ax = df.groupby("Location")["Revenue generated"].sum().plot.bar(color="Orange")

# Add numbers on top of the bars
ax.bar_label(ax.containers[0], fmt="%d", padding=3)

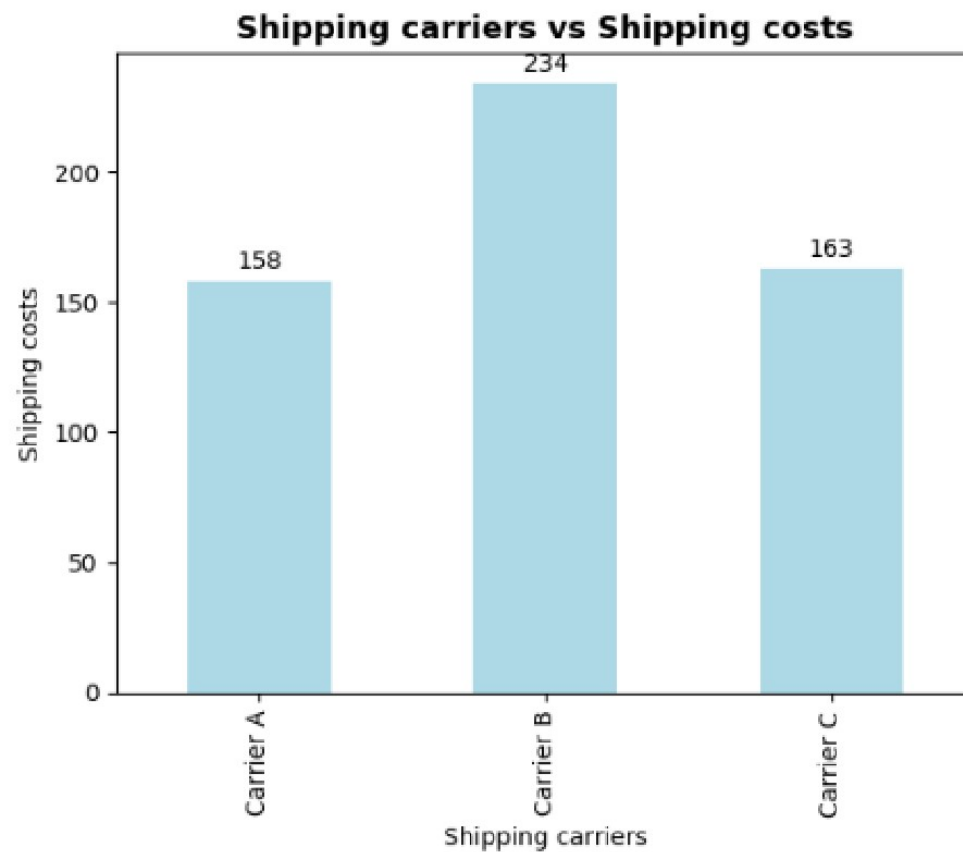
plt.title("Location vs Revenue generated ", fontsize=13, fontweight="bold")
plt.xlabel("Location")
plt.ylabel("Revenue generated")
plt.show()
```



```
[69]: ax = df.groupby("Shipping carriers")["Shipping costs"].sum().plot.bar(color="lightblue")

# Add numbers on top of the bars
ax.bar_label(ax.containers[0], fmt="%d", padding=3)

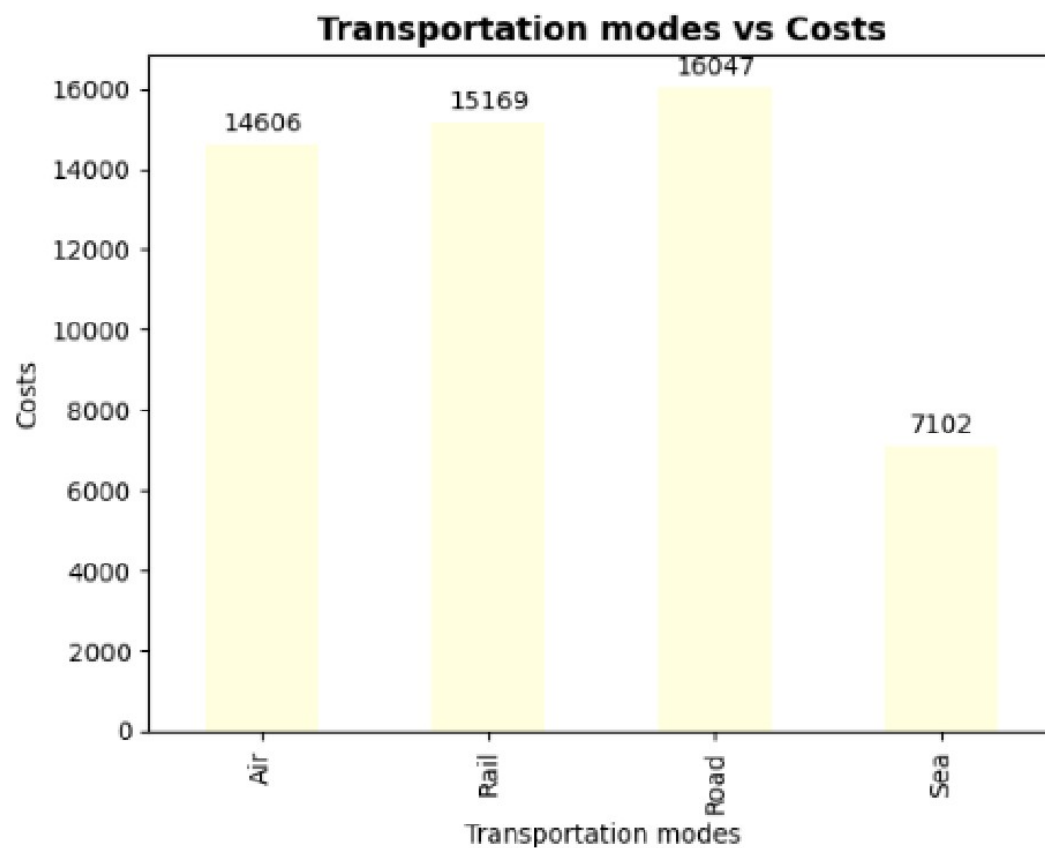
plt.title("Shipping carriers vs Shipping costs", fontsize=13, fontweight="bold")
plt.xlabel("Shipping carriers")
plt.ylabel("Shipping costs")
plt.show()
```



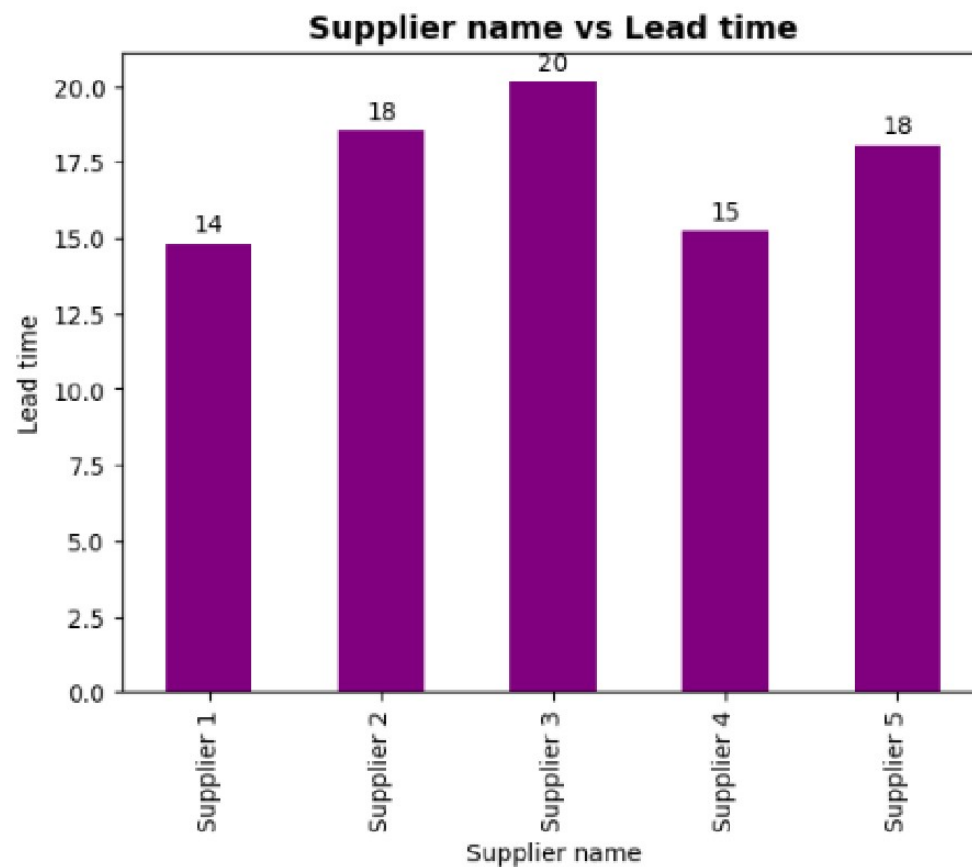
```
[70]: ax = df.groupby("Transportation modes")["Costs"].sum().plot.bar(color="lightyellow")

# Add numbers on top of the bars
ax.bar_label(ax.containers[0], fmt="%d", padding=3)

plt.title("Transportation modes vs Costs", fontsize=13, fontweight="bold")
plt.xlabel("Transportation modes")
plt.ylabel("Costs")
plt.show()
```



```
[75]: ax = df.groupby("Supplier name")["Lead time"].mean().plot.bar(color="Purple")
ax.bar_label(ax.containers[0], fmt="%d", padding=3)
plt.title("Supplier name vs Lead time", fontsize=13, fontweight="bold")
plt.xlabel("Supplier name")
plt.ylabel("Lead time")
plt.show()
```



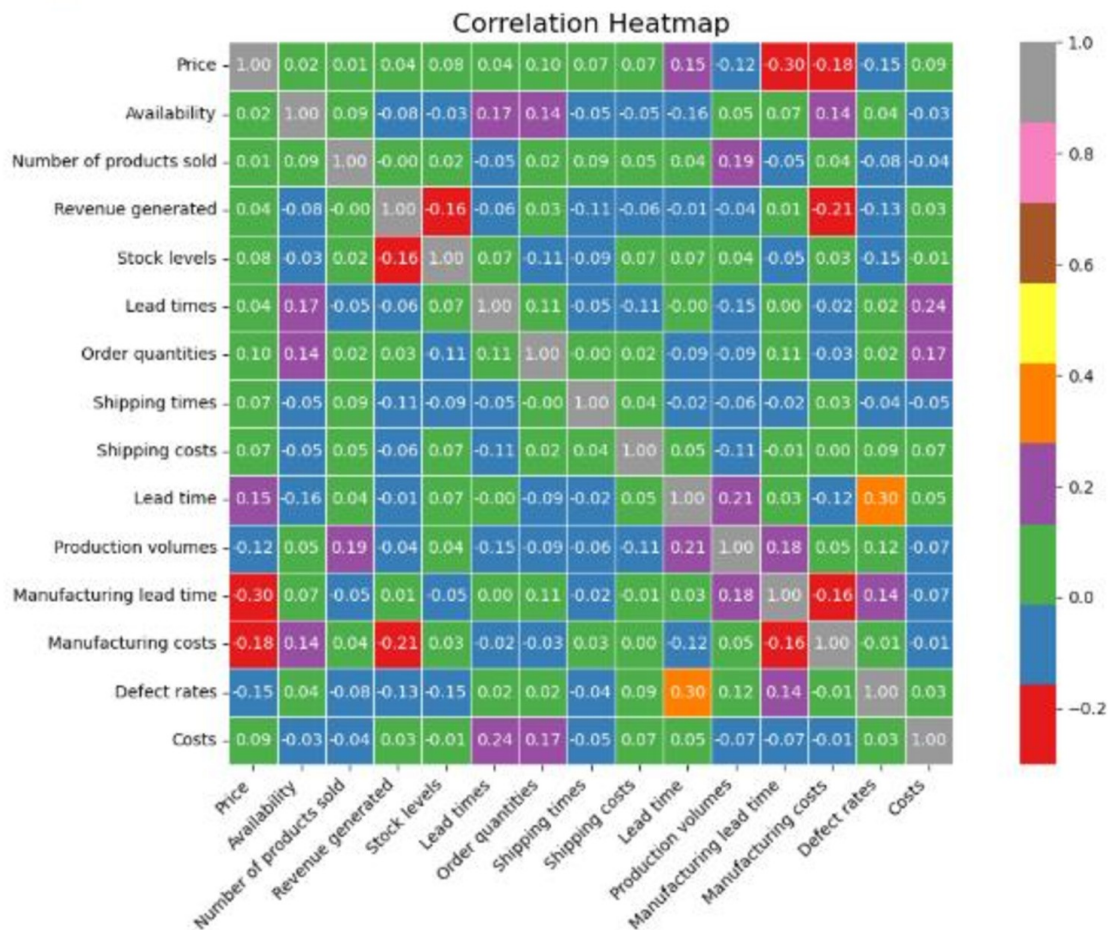
```
[77]: import matplotlib.pyplot as plt
import seaborn as sns

# Calculate correlation
correlation = df.corr(numeric_only=True)

# Create heatmap
plt.figure(figsize=(12, 8))

sns.heatmap(
    correlation,
    annot=True,
    fmt='.2f',
    cmap='Set1',
    linewidths=0.5,
    square=True
)

plt.title("Correlation Heatmap", fontsize=16)
plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()
```



SUPPLY CHAIN ANALYSIS

Total Revenue
25K

Average Ship cost
15

Total Units Sold
2K

Average Defect Rate
1.83

Total Revenue
by Location

16123

Total Units Sold
by Product type

736

Average Defect Rate
by Transportation modes

1.83

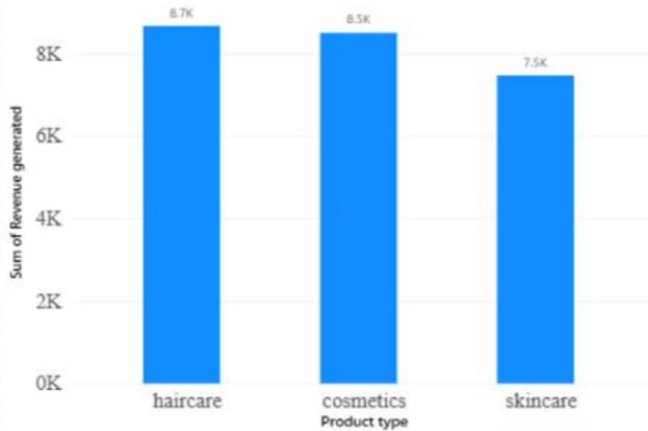
Product type

- ☐ Select all
- ☐ cosmetics
- ☐ haircare
- ☐ skincare

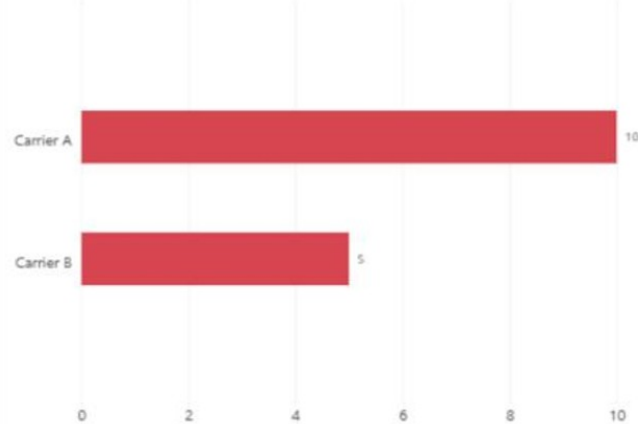
Location

- ☐ Select all
- ☐ Bangalore
- ☐ Mumbai

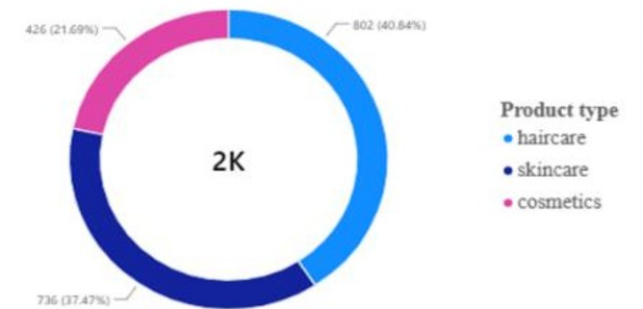
Sum of Revenue generated
by Product type



Sum of Shipping costs
by Shipping carriers



Sum of Number of products sold
by Product type



Product type	SKU	Price	Availability	Number of products sold	Revenue generated	Stock levels	Lead times	Order quantities	Shipping times	Shipping carriers	Shipping costs	Supplier name	Location
haircare	SKU0	70	55	802	8662	58	7	96	4	Carrier B	3	Supplier 3	Mumbai
skincare	SKU1	15	95	736	7461	53	30	37	2	Carrier A	10	Supplier 3	Mumbai
cosmetics	SKU7	43	59	426	8496	93	17	11	1	Carrier B	2	Supplier 4	Bangalore



Product type ▼

Revenue generated ▼

haircare

8662

skincare

7461

cosmetics

8496



Product type



Number of products sold



haircare

802

skincare

736

cosmetics

426

 DAX queries will be saved to your model. They won't be visible when published in the Power BI service. [Learn more](#) 



 Run

 Update model with changes (0)

```
1  EVALUATE
2    SUMMARIZECOLUMNS(
3      |   "Total Revenue", [Total Revenue]
4    )
```

Results

Result 1 of 1 

 Copy 



	[Total Revenue]
1	24619

 DAX queries will be saved to your model. They won't be visible when published in the Power BI service. [Learn more](#) 

 Run  Update model with changes (0)

```
1 EVALUATE
2   SUMMARIZECOLUMNS(
3     "Total Units Sold", [Total Units Sold]
4   )
```

Results | Result 1 of 1  Copy 

	[Total Units Sold]
1	1964

 DAX queries will be saved to your model. They won't be visible when published in the Power BI service. [Learn more](#) 

 Run

 Update model with changes (0)

```
1  EVALUATE
2  |  SUMMARIZECOLUMNS(
3  |    "Average Ship cost", [Average Ship cost]
4  |  )
```

Results

Result 1 of 1 

 Copy 

	[Average Ship cost]
1	15

 DAX queries will be saved to your model. They won't be visible when published in the Power BI service. [Learn more](#) 

 Run

 Update model with changes (0)

```
1 EVALUATE
2     SUMMARIZECOLUMNS(
3         "Average Defect Rate", [Average Defect Rate]
4     )
```

Results

Result 1 of 1 

 Copy 

	[Average Defect Rate]
1	1.83